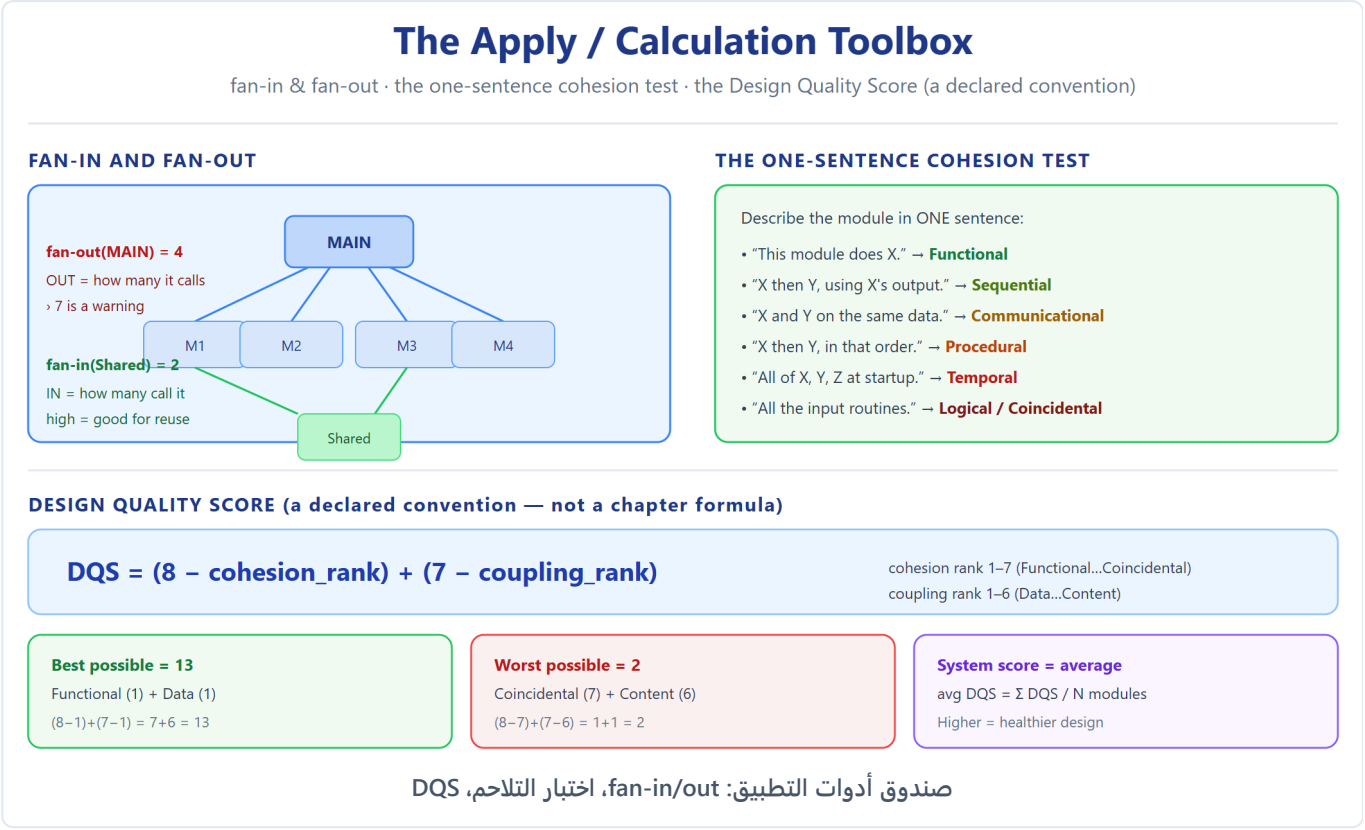
وثيقة المراجعة والتلخيص الأكاديمي

ASE · Chapter 5 · Software Design — Aggarwal & Singh, 3rd ed. (Bloom Level 3 — Apply)

المصدر: سلايدات Ch05_Software_Design.pdf (1–99) + book p.202–251. Fig. 4 = Modularity and software cost؛ Fig. 7 (slide 19) = ترتيب الاقتران؛ Fig. 11 (slide 27) = ترتيب التلاحم؛ سلايدات 73–80 = خطوات OOD الست. **الهدف:** تُعطى حالة ملموسة (رقم، مخطط، وصف واجهة)، فتطبّق القاعدة وتحسب النتيجة خطوة بخطوة. **قاعدتان ثابتتان (لا تُخالفان):** الاقتران الأفضل ← الأسوأ = **Data < Stamp < Control < External < Common < Content** (Fig 7، slide 19) — 6 أنواع. التلاحم الأفضل ← الأسوأ = **Functional < Sequential < Communicational < Procedural < Temporal < Logical < Coincidental** (Fig 11، slide 27) — 7 أنواع. **ملاحظة استعمال:** سؤال إنكليزي، جواب إنكليزي كامل بخطوات الحساب، تبرير عربي قصير في النهاية. أي ترتيب عددي (rank) مخترع للحساب يُعلَن صراحةً أنه **اصطلاح للتمرين**، مو نص الفصل.

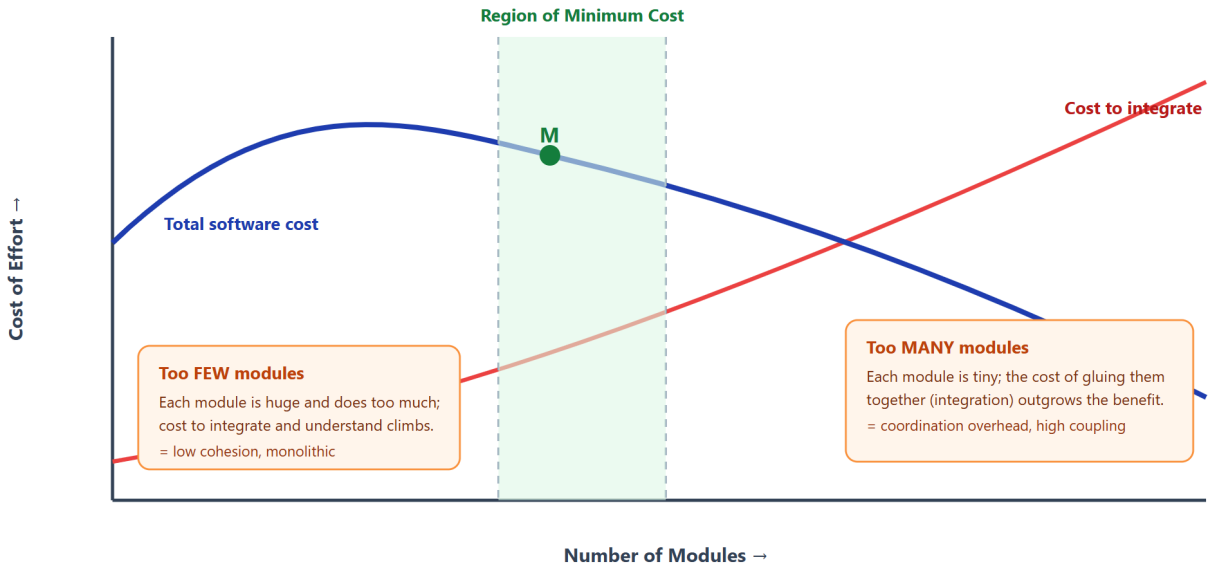
الرسوم المرجعية (راجعها قبل الحل)



صندوق أدوات التطبيق: fan-in/out، اختبار التلاحم، DQS

Modularity and Software Cost

Fig. 4 (slide 14) · there is an optimum number of modules — not "as many as possible"



Takeaway: stop at point M — the design that minimises total cost, not the design with the most (or fewest) modules.

منحنى الوحدات والتكلفة (Region of Minimum Cost)

قاعدة العمل (Marking Conventions Declared Up-Front)

Convention 1 — Coupling rank (my convention for scoring only, NOT a chapter fact):

Coupling type	Rank (1 = best)
Data	1
Stamp	2
Control	3
External	4
Common	5
Content	6

Convention 2 — Cohesion rank (my convention for scoring only, NOT a chapter fact):

Cohesion type	Rank (1 = best)
Functional	1
Sequential	2
Communicational	3

Cohesion type	Rank (1 = best)
Procedural	4
Temporal	5
Logical	6
Coincidental	7

Convention 3 — Design Quality Score (DQS) — my convention: For one module, DQS = (8 − cohesion_rank) + (7 − coupling_rank). Range = 2 (worst possible: Coincidental L7 + Content C6) to 13 (best possible: Functional L1 + Data C1). Higher = better. For a system of N modules, Average DQS = Σ DQS / N.

تذكير: الترتيبين 1–6 و1–7 مطابقان للـ VERIFIED FACTS؛ الأرقام مجرد تسمية رتبة، الترتيب نفسه هو المهم. الـ DQS **اصطلاح مخترع للتمرين**، مو من الفصل.

القسم أ — حساب Fan-in / Fan-out على Structure Chart

Q7.1 (Fan-out / fan-in computation — the call graph)

> A structure chart has these calls: - M1 calls M2, M3, M4 - M2 calls M5 - M3 calls M5, M6, M7, M8, M9, M10, M11, M12 - M4 calls M6 - M5, M6, M7–M12 call nothing.

Task: Compute fan-out for every module. Flag any module exceeding 7. Compute fan-in for M5, M6, M3. Identify the highest-reuse module by fan-in.

Worked answer — Step 1 (fan-out = number of modules a module calls):

Module	Calls	fan-out
M1	M2, M3, M4	**3**
M2	M5	**1**
M3	M5, M6, M7, M8, M9, M10, M11, M12	**8**
M4	M6	**1**
M5	—	**0**
M6	—	**0**
M7–M12	—	0 each

Step 2 (flag > 7): M3 = 8 → **exceeds 7** → **warning sign** (module doing too much; candidate for splitting). M1 = 3 → acceptable.

Step 3 (fan-in = number of modules that call a module):

Fan-in(M5): called by M2 and M3 → **2**
Fan-in(M6): called by M3 and M4 → **2**
Fan-in(M3): called by M1 → **1**

Step 4 (second pass — re-check the arithmetic): M3's list: M5, M6, M7, M8, M9, M10, M11, M12 → count = 8 ✓. M5 callers = {M2, M3} = 2 ✓. M6 callers = {M3, M4} = 2 ✓.

Step 5 (highest reuse): M5 and M6 tie at fan-in = 2 → **both are the highest-reuse modules**; they are shared leaves and must carry the cleanest interfaces (a change here ripples to 2 callers each).

> = high fan-in" rule and "warning = 7 < fan-out" The **[Foundational Knowledge / Standard Concept]** The deck's structure-chart .**NOT stated in the Ch05 deck** ,*standard SE rules of thumb* are "good for reuse .define the notation only. Flagged every time (43–42 سلايدات ,18–Fig 17) slides

التبرير العربي: الـ fan-out = عدد الوحدات اللي تستدعيها الوحدة، والـ fan-in = عدد الوحدات اللي تستدعيها. M3 عندها fan-out = 8 وهذا يتجاوز الحد الشائع (7) فيكون إشارة خطر: الوحدة تتحكم بزائد عن اللزوم وتُرشّح للتقسيم. M5 و M6 عندهم أعلى fan-in (=2) فهُم وحدات إعادة الاستخدام، وأي تغيير فيهم ينتشر للوحدتين. القاعدة (fan-out<7) قاعدة متعارف عليها **مو من الفصل** — موسمة.

Q7.2 (Fan-out computation — the wide caller, second instance)

> A module main_controller( ) directly calls: read_config, validate_input, parse_record, store_record, log_event, update_index, notify_user, refresh_display, write_audit, close_session — ten calls. Modules read_config and close_session are also called by session_manager( ).

Task: State main_controller's fan-out. Flag it. Compute fan-in of read_config.

Worked answer:

Step 1: fan-out(main_controller) = count the calls = 10.

Step 2: 10 > 7 → **warning sign**. A module that fans out to 10 subordinates is controlling too many distinct concerns; the standard remedy is to introduce an intermediate controller and redistribute.

Step 3 (re-check): read_config, validate_input, parse_record, store_record, log_event, update_index, notify_user, refresh_display, write_audit, close_session → count = 10 ✓.

Step 4: fan-in(read_config) = called by main_controller and session_manager → **2**.

التبرير العربي: fan-out = 10 > 7 ⇒ تحذير: الوحدة توزّع السيطرة على عشر مهام مختلفة، وهذا يعني انخفاض تماسكها وارتفاع احتمال التغيير فيها. الحل: إدخال وحدة وسيطة توزّع النداءات. fan-in للـ read_config = 2 فتكون وحدة مشتركة تستحق واجهة نظيفة. (القاعدة موسمة كـ standard concept.)

Q7.3 (Fan-in / fan-out — the reuse hot-spot with a trap)

> Call graph: - A calls B, C - B calls D - C calls D, E - D calls F - E calls F - F calls nothing - G (a second root) calls D

Task: Compute fan-out for every module; compute fan-in for D and F; identify which module is (i) the most-reused and (ii) the highest ripple risk. Flag any fan-out breach.

Worked answer:

Module	Calls	fan-out
A	B, C	2
B	D	1
C	D, E	2
D	F	1
E	F	1
F	—	0
G	D	1

Fan-in:

Fan-in(D): called by B, C, G → **3**
Fan-in(F): called by D, E → **2**

Re-check: D callers = {B, C, G} = 3 ✓. F callers = {D, E} = 2 ✓. No fan-out exceeds 7 (max = 2) → **no breach**.

(i) Most reused: D, fan-in = 3. **(ii) Highest ripple risk:** D again — it is both high fan-in (3 dependents) *and* it sits on the path to F, so a defect in D propagates to B, C, G and then to F and its dependents. D is the critical node.

التبرير العربي: لا يوجد fan-out > 7 فالبنية سليمة من ناحية اتساع السيطرة. لكن D هي نقطة الخطر الحقيقية: أعلى fan-in (=3) وفي نفس الوقت تقع على مسار الوصول لـ F، فأي تغيير فيها ينتشر لكل مستدعييها ثم إلى F. أعلى fan-in = أعلى إعادة استخدام وأعلى مخاطرة ripple في نفس الوقت. (standard concept = fan-in rule، موسمة.)

القسم ب — تصنيف وحدات + حساب درجة جودة التصميم (DQS)

Q7.4 (Classify N modules + compute the Design Quality Score)

> Four modules are described by their interface and internal structure: - **P** compute_tax(salary) — one self-contained calculation; called only with the primitive salary. **Interface: single primitive parameter.** - **Q** validate_then_store(record) — validates record, then the validated record feeds store() inside the same module. **Interface: one object passed; internals form an output → input chain.** - **R** run_startup() — opens the log, initialises drivers, sets the time zone. All touch a shared global SYSTEM_STATE. **Interface: no parameters; all through the global.** - **S** handle_request(req, mode, flags) — a dispatcher that reads mode to decide which of six tasks to run; it also reads and writes another module's internal buffer directly.

Task: Classify cohesion and coupling for each, then compute the DQS per module and the average DQS for the system.

Worked answer — Step 1 (classify, using slides 19 & 27):

Module	Cohesion	Cohesion rank	Coupling	Coupling rank
P	Functional	1	Data	1
Q	Sequential	2	Stamp	2
R	Temporal	5	Common	5
S	Logical	6	Content	6

P: single task ("compute tax") → **Functional**. Only a primitive passed → **Data**.
Q: validate → store is an output → input chain (slide 28) → **Sequential**. A whole record object is passed though only validation fields are needed → **Stamp**.
R: elements grouped because they run at the same startup time → **Temporal**. All share a global SYSTEM_STATE → **Common**.
S: six tasks chosen by mode at runtime → **Logical**; mode/flags are control info → **Control** at the interface, **but** it also reaches into another module's internal buffer → **Content** (worse of the two; content dominates). Content rank = 6.

Step 2 (compute DQS = (8 − cohesion_rank) + (7 − coupling_rank)):

Module	Cohesion term (8 − r)	Coupling term (7 − r)	DQS
P	8 − 1 = 7	7 − 1 = 6	**13**
Q	8 − 2 = 6	7 − 2 = 5	**11**
R	8 − 5 = 3	7 − 5 = 2	**5**
S	8 − 6 = 2	7 − 6 = 1	**3**

Step 3 (average): (13 + 11 + 5 + 3) / 4 = 32 / 4 = **8.0**.

Step 4 (re-check all arithmetic): P: 8−1=7, 7−1=6, 7+6=13 ✓. Q: 8−2=6, 7−2=5, 6+5=11 ✓. R: 8−5=3, 7−5=2, 3+2=5 ✓. S: 8−6=2, 7−6=1, 2+1=3 ✓. Sum 13+11+5+3 = 32; 32/4 = 8.0 ✓.

Step 5 (interpretation): P (13) and Q (11) are healthy; R (5) is weak; S (3) is the system's weakest link — lowest on **both** axes. **Refactor S first**: split the six logical tasks into single-purpose modules (raises cohesion) and replace the internal-buffer access with a data parameter (removes the worst coupling). Moving S from 3 toward ~11–13 is worth far more than polishing P or Q. R next: pass config values as parameters instead of using the global.

> **التنبيه:** الـ DQS **اصطلاح مخترع لهذا التمرين** — مو من الفصل. الفصل يعطي الترتيب فقط (Fig 7, Fig 11)، والأرقام هنا تحويل ذلك الترتيب إلى سلم نقاط.

التبرير العربي: P و Q سليمتان (functional/data و sequential/stamp). R سيئة بشكل متوسط (temporal/common). S هي الأسوأ: logical cohesion + content coupling معًا، فتأخذ أدنى نقطة (3). الترتيب الرقمي 1–6 للاقتران و1–7 للتلاحم مطابق للفصل؛ أما DQS فهو اصطلاح التمرين. أول إصلاح = S؛ لأن تحسين الأسوأ يعطي أكبر مكسب بأقل جهد.

Q7.5 (Classify + score a five-module design, second instance)

- **T1** sort_employees(list， ascending_flag) — sorts; the flag chooses the order. Passes only the list and flag.
- **T2** print_all_reports() — prints the payroll report, the attendance report, the tax report; grouped because all are printing tasks.
- **T3** read_sensor()； scale_reading()； store_reading() — each step feeds the next; interface passes one scaled value.
- **T4** update_ledger() — writes into the ACCOUNTS global table shared by 5 other modules.
- **T5** misc_helpers() — 30 unrelated helper routines; also directly edits the ACCOUNTS global.

Task: Classify, compute DQS per module, compute average, and rank the three most urgent refactors.

Worked answer — Step 1:

Module	Cohesion	rank	Coupling	rank
T1	Logical (flag selects behaviour)	6	Control (flag is control info)	3
T2	Logical (same logical class: printing)	6	Data (only data passed)	1
T3	Sequential	2	Data	1
T4	Functional (one task)	1	Common (shared global)	5
T5	Coincidental	7	Content (edits another's internals)	6

T1: the ascending_flag selects *which behaviour* → **Logical** cohesion (runtime-selected tasks) and **Control** coupling (control information passed).
T2: three printing routines grouped by "same logical class" → **Logical** (not coincidental — the printing class is a real, nameable relation) but only data is passed → **Data**.
T3: read → scale → store chain → **Sequential**; single value passed → **Data**.
T4: single task → **Functional**; but shares ACCOUNTS global → **Common**.
T5: 30 unrelated helpers → **Coincidental**; edits ACCOUNTS internals → **Content**.

Step 2 (DQS):

Module	Cohesion term	Coupling term	DQS
T1	8 − 6 = 2	7 − 3 = 4	**6**
T2	8 − 6 = 2	7 − 1 = 6	**8**
T3	8 − 2 = 6	7 − 1 = 6	**12**
T4	8 − 1 = 7	7 − 5 = 2	**9**
T5	8 − 7 = 1	7 − 6 = 1	**2**

Step 3 (average): (6 + 8 + 12 + 9 + 2) / 5 = 37 / 5 = **7.4**.

Step 4 (re-check): T1: 2+4=6 ✓. T2: 2+6=8 ✓. T3: 6+6=12 ✓. T4: 7+2=9 ✓. T5: 1+1=2 ✓. Sum 6+8+12+9+2 = 37; 37/5 = 7.4 ✓.

Step 5 (three most urgent refactors): 1. **T5** (DQS 2) — worst on both axes; split the 30 helpers by function and remove the internal ACCOUNTS access → content → data. 2. **T4** (Common, DQS 9 but *coupling* worst-but-one) — high cohesion is undercut by a shared global; pass needed values as parameters → common → data. 3. **T1** (DQS 6) — split the flag-selected behaviour into two single-purpose routines → logical → functional, control → data.

التبرير العربي: T3 هي الأنظف (12). T5 هي الأسوأ بلا منافس (2) لأنها content + coincidental — إصلاحها أولًا. T4 تلاحمها ممتاز لكن الاقتران common يخفضها ثانيًا. T1 ثالثًا لأن الـ control coupling = flag وتلاحم logical. الأرقام (DQS) اصطلاح التمرين، والترتيب مطابق للفصل.

القسم ج — Region of Minimum Cost (منحنى المعيارية–الكلفة)

Q7.6 (Locate the Region of Minimum Cost — read the U-curve)

> A team estimates the following total cost (in developer-days) for different numbers of modules in the same system (curve shape from Fig. 4, slide 14):
>
> | Modules | Total cost (days) |
> |---|---|
> | 1 | 420 |
> | 3 | 300 |
> | 5 | 230 |
> | 7 | 195 |
> | **9** | **175** |
> | 11 | 185 |
> | 15 | 260 |
> | 20 | 400 |

Task: Locate the Region of Minimum Cost. Explain the two failure modes on either side.

Worked answer — Step 1 (find the minimum): Scan the cost column: 420 → 300 → 230 → 195 → **175 (min)** → 185 → 260 → 400. Minimum at **9 modules, cost = 175 days**.

Step 2 (the region, not a single point): The costs at 7 (195) and 11 (185) are close to 175, so the practical **Region of Minimum Cost ≈ 7–11 modules**, centred on 9. Outside this band the cost rises steeply.

Step 3 (re-check): ascending to the min: 420, 300, 230, 195, 175 → strictly decreasing to 9 ✓; after 9: 185, 260, 400 → strictly increasing ✓. Genuine U-shape confirmed. 175 is the smallest value in the column ✓.

Step 4 (left failure mode — too few modules): At 1–3 modules the cost is high because the whole system is one indissoluble block: the cost of **integration and communication is minimal, but the cost of developing/understanding/maintaining monolithic units explodes** — the single module cannot be separated for testing, and every change is global. The curve's left branch is driven by **module development effort** (a big, unmanageable unit costs more to build and change). Read side: "too few" = poor modularity = high development/maintenance cost.

Step 5 (right failure mode — too many modules): At 15–20 modules the cost rises again because more modules means **more interfaces, more integration effort, more communication overhead** between modules. Each additional boundary must be designed, documented and integrated. The right branch is driven by **integration/interface overhead**. Read side: "too many" = integration cost dominates.

Step 6 (significance): The minimum exists because the two cost components move in opposite directions — as you add modules, *development* cost falls but *integration* cost rises. Where the sum is lowest is the Region of Minimum Cost.

التبرير العربي: منحنى Fig. 4 (سلايد 14) على شكل U: كل ما زادت الوحدات تنخفض كلفة التطوير لكن ترتفع كلفة التكامل بينها. المنطقة الأقل كلفة هنا ≈ 7–11 وحدة (الحد الأدنى عند 9). على اليسار (وحدات قليلة جدًا) الكلفة عالية لأن الوحدة العملاقة غير قابلة للفهم/الاختبار. على اليمين (وحدات كثيرة جدًا) الكلفة عالية بسبب كلفة الواجهات والتكامل. الأمثل عند تقاطع النزعتين.

Q7.7 (Read the U-curve — second instance, asymmetric curve)

Another estimate:

Modules	Total cost (days)
2	380
4	250
6	180
8	150
10	145
12	160
16	220
24	390

Task: Locate the minimum and the region; state on which side this curve is steeper and what that implies.

Worked answer — Step 1: Minimum = **145 days at 10 modules** (next lowest: 150 at 8, 160 at 12).

Step 2 (region): 8–12 modules ≈ Region of Minimum Cost (costs 145–160).

Step 3 (re-check): 380 → 250 → 180 → 150 → 145 decreasing to 10 ✓; after 10: 160 → 220 → 390 increasing ✓. Min value 145 ✓.

Step 4 (steepness): From 10 → 24 modules the cost climbs 145 → 390 (+245). From 10 → 2 modules (going left) the cost climbs 145 → 380 (+235). The two sides are close; but the **right-hand jump from 12→24 is +230 over 12 added modules, while the left from 8→2 is +230 over 6 removed modules** — per-module, the **left side is steeper**. Implication: in this system, *under-modularising* (too few, giant modules) is punished more per unit than over-modularising — a sign the modules are individually very large, so merging them is especially costly.

التبرير العربي: الحد الأدنى عند 10 وحدات (145 يوم)، والمنطقة الاقتصادية ≈ 8–12. الجانب الأيسر أكثر حدّة لكل وحدة: تقليل عدد الوحدات (دمج وحدات كبيرة) يعاقَب بقوة أكبر، وهذا يعني أن الوحدات الأساسية ضخمة جدًا ودمجها مكلف. الجانبان يمثّلان نمطي الفشل: وحدات قليلة جدًا (كلفة فهم/صيانة) أو كثيرة جدًا (كلفة واجهات/تكامل).

القسم د — One-Sentence Cohesion Test على 6 دوال

Q7.8 (Apply the one-sentence test to six routines)

> **The one-sentence test:** Read the routine and try to describe what it does in exactly one sentence with **one verb**. If you succeed, cohesion is high (Functional). If the best sentence needs "and", "or", a flag-word ("depending on…"), "first…then…", or "all…at the same time", note which connector you needed — the connector names the cohesion type.
>
> Classify these six routines:

computeGrossPay(hours, rate) — "computes gross pay from hours and rate".
openFiles(); loadTables(); initialisePrinter(); grouped as systemStartup() — "does open files **and** load tables **and** init printer, all at startup".
readRaw(); cleanRaw(); formatOutput(); — "reads raw data, **then** cleans it, **then** formats it".
saveEmployee(rec) and printEmployee(rec) grouped as employeeIO(rec) — "saves the employee **or** prints the employee depending on the operation flag".
sortNames(); computeAverage(); lookupId(); grouped as miscArrayOps() — "sorts names **and** computes average **and** looks up an id".
validateHeader(); printHeader(); storeHeader(); all operating on the **same** input headerRecord — "takes the header record and validates it, prints it, and stores it".

Worked answer:

#	Routine(s)	Connector needed in the one sentence	Cohesion	rank
1	computeGrossPay	none — one verb	**Functional**	1
2	systemStartup	"…**and**…, **all at the same time** (startup)"	**Temporal**	5
3	read → clean → format	"…**then**…" (output → input chain)	**Sequential**	2
4	employeeIO(rec)	"…**or**…depending on the flag"	**Logical**	6
5	miscArrayOps	"…**and**…" with no unifying relation	**Coincidental**	7

#	Routine(s)	Connector needed in the one sentence	Cohesion	rank
6	header trio on one input	none of the chain words; all consume the **same input**	**Communicational**	3

Step (re-check the two tricky ones): - #3 vs #6: #3 has an **output → input chain** (each step feeds the next) → Sequential. #6 has **independent operations on a shared input**, no chaining → Communicational. Different connectors: "then" vs "same input". ✓ - #2 vs #5: #2 has a real reason ("all at startup") → Temporal (rank 5). #5 has no reason at all → Coincidental (rank 7). ✓ - #4: the word "or / depending on the flag" is the classic Logical-cohesion marker (runtime task selection) → rank 6. ✓

التبرير العربي: امتحان الجملة الواحدة يكشف نوع التلاحم من الرابط (connector) المطلوب: بدون أي رابط ⇒ Functional (1). "ثم" ⇒ Sequential (3). "نفس المدخل" ⇒ Communicational (6). "أو / حسب الـ flag" ⇒ Logical (4). "كلها وقت واحد/startup" ⇒ Temporal (2). لا يوجد سبب على الإطلاق ⇒ Coincidental (5). الأرقام = رتبة على سلم Fig 11 (slide 27).

القسم هـ — OOD 6-Step Processes

Q7.9 (Apply the OOD 6-step process — output the artefact list)

> Problem: "A **Library Book-Loan** system must let a Librarian issue a book to a member and let the member return it."

Task: Apply the six OOD steps (slides 73–80) and output, for **each** step, the artefact/diagram or decision produced. Do not draw the artefacts — list them.

Worked answer (six steps, quoting the deck's step names):

Step	Deck name (slides 73–80)	Artefact / decision produced for this problem
1	i. Create use case model	Actor list (**Librarian, Member**); use cases **"Issue Book"**, **"Return Book"**; a **use-case diagram** linking actors to those use cases.
2	ii. Draw activity diagram *(if required)*	**Activity diagram** for the flow "Issue Book": request → verify membership → check availability → record loan → confirm — a flow of control from activity to activity.
3	iii. Draw the interaction diagram	Object list per use case (**LibrarianUI, LoanController, Book, Loan** — interface/control/entity objects); **sequence diagrams** (and collaboration diagrams) for each use case showing messages.
4	iv. Draw the class diagram	**Class diagram**: classes Book, Member, Loan, Librarian with attributes and operations; relationships — **association** (Member–Loan), **aggregation** (Loan contains Book), **dependency**, **generalisation** (if a ReferenceBook/LendingBook split is chosen).
5	v. Design of state chart diagrams *(if required)*	**State chart diagram** for Book: Available → Issued → Returned → Available, with the events that cause each transition.
6	vi. Draw component and development (deployment) diagram	**Component diagram** (implementation view: which classes map to which components) + **deployment diagram** (physical distribution: library terminals/PCs and the DB server) → the **design documents**.

and flagged (state chart ,activity) 2 = in the deck "if required" steps marked ;✓ 6 = steps listed **:(re-check) الإحصاء** ,aggregation ,dependency ,association) **four** the deck's = class-diagram relationships named ;✓ as optional here .✓ (generalisation

التبرير العربي: خطوات OOD الست مطابقة للفصل (سلايدات 73–80): use case ← activity ← interaction ← class ← state chart. component/deployment خطوتان موسومتان "if required" بالكتاب (activity و state chart) فتبقى اختياريتين. المطلوب هنا قائمة المخرجات لكل خطوة، مو رسمها. خطوة الـ class diagram تحديدًا تنتج الـ 4 علاقات المذكورة بالفصل.

Q7.10 (Apply the OOD 6-step process — second problem, verify the ordering)

> Problem: "A **Parking Lot** system assigns a free spot to an arriving vehicle, computes the fee on exit, and reports the number of free spots."

Task: For each of the six steps, state the artefact produced AND the single most important **design decision** taken at that step. Confirm the step order matches the deck.

Worked answer:

Step	Artefact	Key decision made here
1 use case model	Actor list (**Driver, Attendant**); use cases *Enter Lot*, *Exit Lot*, *Check Availability*; use-case diagram	Fix the **system boundary** and actors (decides scope).
2 activity diagram	Activity diagram for *Enter Lot* (assign spot → raise barrier → record ticket)	The **ordering** of activities (dynamic flow).
3 interaction diagram	Objects Barrier， Ticket， ParkingSpot， FeeCalculator; sequence diagram per use case	The **messages** and therefore the intended **coupling** between objects.
4 class diagram	Class diagram: Vehicle， ParkingSpot， Ticket， FeeCalculator; relationships: **aggregation** (Lot aggregates ParkingSpot), **association** (Ticket–Vehicle), **dependency** (FeeCalculator depends on Ticket), **generalisation** (Car/Truck/Motorcycle → Vehicle)	The **state (attributes) and behaviour (operations)** plus the four relationship kinds.
5 state chart	State chart for ParkingSpot: Free → Occupied → Free	The **state transitions** that the class must support.
6 component/deployment	Component diagram + deployment diagram (kiosk PC, server, barrier controller)	The **physical mapping** of components to hardware.

Order check: 1 → 2 → 3 → 4 → 5 → 6, exactly as the deck presents them (use case → activity → interaction → class → state chart → component/deployment). ✓

التبرير العربي: نفس الخطوات الست، وكل خطوة تقيّد اللي بعدها: الـ use case يثبّت الحدود، الـ activity يثبّت الترتيب، الـ interaction يكشف الرسائل/الاقتران، الـ class يثبّت الحالة والسلوك و4 العلاقات، الـ state chart يثبّت الانتقالات، والـ component/deployment يثبّت التوزيع الفيزيائي. الترتيب مطابق للفصل (سلايدات 73–80).

القسم و — تتبّع أثر التغيير: Data vs Content (عدّ وإحصاء)

Q7.11 (Trace and COUNT which modules must be re-checked)

> A design has these modules and links: - **Entry** calls **Save**, **Print** - **Save** calls **Format**, **Log** - **Print** calls **Format** - **Format** calls **Util** - **Log** calls **Util**
>
> Two scenarios for the interface between Save and Format: - **(A) Data coupling:** Save passes Format(record) — one object, and Format uses only the fields it receives. Format's internals are private. - **(B) Content coupling:** Save writes directly into Format's internal buffer Format.buf and reads Format.lastError.

Task: A change is made **inside Format** (its internal buffer layout changes). Under (A) and under (B), enumerate exactly which modules must be re-checked. Give the count for each.

Worked answer — Step 1 (build the dependency directions):

Under **(A) data coupling:** Save depends on Format **only through the parameter record**. The callers of Format are Save and Print; callers of those are Entry. But the interface contract is *only* the record parameter, and the change is **internal** to Format → the existing interfaces still hold.

Modules to re-check under (A): Format itself only. The interface with Save/Print is unchanged (same parameter), so **no caller needs re-checking**. **Count (A) = 1** (Format).

Under **(B) content coupling:** Save reaches into Format.buf and Format.lastError. Changing Format's internal buffer layout **breaks Save directly**, because Save depends on those internals. Save's callers (Entry) may then see changed behaviour and must be re-checked too, as must anything Save feeds. Walk the graph:

Format — changed (source).

Save — reads/writes Format.buf and Format.lastError → **must re-check** (its code assumes the old layout).
Entry — calls Save; if Save's behaviour/return changes, Entry must be re-checked.
Print — calls Format; Print itself does not touch Format.buf, but if the *observable* behaviour of Format changed, Print must be re-verified against the new behaviour.
Log and Util — Util is downstream of both Save and Log; the risk path does not originate from them, so they need re-check only if Format's public output changed. Conservative count includes them as *verification candidates*.

Count (B), strict (interface-breaking chain): Format, Save, Entry = **3**. **Count (B), conservative (all dependents that could observe a behaviour change):** Format, Save, Entry, Print, Log, Util = **6**.

Step 2 (re-check the counts by re-walking): - (A): change is internal; contract unchanged → only Format. Count = **1** ✓. - (B) strict: the *content* link is Save ↔ Format; the modules on that inward path are Save and (its caller) Entry; plus the changed module Format → 3 ✓. - (B) conservative: add every module reachable from

Format in the call graph = Save, Print, Entry, Log, Util + Format → 6 ✓ (7 modules total in the design; only none is truly unaffected).

Step 3 (the lesson): Same graph, same change — the **coupling form** decides the blast radius. Data coupling contains the ripple to **1** module; content coupling spreads re-checking to **3 (minimum) to 6 (conservative)**. That factor-of-3-to-6 difference is the quantitative face of "content coupling is the worst".

التبرير العربي: نفس التغيير (داخل Format) ونفس المخطط، لكن الاقتران يحدد حجم الانتشار: بالـ data coupling التغيير داخلي والعقد (البارامتر) ما تغيّر ⇒ نراجع وحدة واحدة (1). بالـ content coupling لأن Save تشتغل مباشرة على buf الداخلية، التغيير يكسرها ⇒ على الأقل 3 وحدات (Format, Save, Entry) وستة بالعدّ المتحفظ (تضيف Print, Log, Util). هذا هو الفرق الكمّي اللي يثبت أن content coupling هو الأسوأ (Fig 7).

Q7.12 (Counting under Common vs Data — the shared-global re-check count)

> A system has 8 modules: M1…M8. Five of them (M2, M3, M5, M6, M8) read and write a shared global table G. The other three (M1, M4, M7) never touch G.
>
> **(A)** The **layout of G** changes (a field is added). **(B)** Instead, the design is refactored so each module passes exactly the fields it needs as parameters, and G is removed; then a module M5's **internal algorithm** changes.

Task: Count how many modules must be re-checked in each case.

Worked answer — Case (A), Common coupling: Any module touching G may depend on its layout. Count = modules reading/writing G = M2, M3, M5, M6, M8 → **5 modules** must be re-checked. M1, M4, M7 are unaffected.

Re-check: ∩ is exactly the five named ✓. Count = 5 ✓.

Worked answer — Case (B), Data coupling after refactor: Now M5's internal algorithm changes. Modules depend on M5 only through its parameter interface, which is unchanged (internal change). Therefore only modules whose **call interface** is affected need re-check — since the interface did not change, **only M5 must be re-checked**. Count = **1**.

Re-check: a purely internal algorithm change under data coupling → interface unchanged → 1 module ✓.

Comparison: Case (A) = 5 modules; Case (B) = 1 module. Same system size (8), but common coupling multiplies the re-check count by 5, data coupling by 1.

التبرير العربي: بالـ common coupling أي تغيير في G يجبرك تراجع **كل** وحدة تلمس G = 5 وحدات (M2, M3, M5, M6, M8). بعد إزالة G والتمرير بالبارامترات (data coupling)، تغيير داخلي في M5 ما يكسر أي عقد ⇒ وحدة واحدة فقط. الفرق 5 مقابل 1 = الوجه الكمّي لقاعدة "قلّل الاقتران".

Q7.13 (Ripple tracing — count the re-check set for content vs data on a chain)

> A chain: Top → A → B → C. B and C communicate through a shared **external file format** (External coupling). Separately, A reaches into B's internal state (Content coupling).

Task: If C changes its file format, count the re-check set under the External coupling. Then, if the design were changed so B–C used data coupling instead, recount. Explain the difference.

Worked answer — Step 1 (External coupling, format of C changes): External coupling means B is tied to C through the **external file format**. Changing that format breaks B. Then B's behaviour may change → A, then Top, must be re-checked.

Re-check set = C (source) + B (external link broken) + A + Top = **4 modules**. **Re-check count:** C, B, A, Top = 4 ✓.

Step 2 (if B–C becomes data coupling): Now B calls C with parameters; the change is *inside* C and C's interface is stable (say C returns the same typed value). Then: - C re-checked (source). - B — interface unchanged (same return) → only needs a light **re-verify**, effectively 0 forced changes; strictly, **no forced re-check** since the contract holds.

Conservative re-check set = C (+ optional verification of B) → **1 (strict) to 2 (conservative)**. **Re-check count:** 1–2 ✓.

Step 3 (re-check the counts): External case: 4 named modules ✓. Data case: contract preserved → 1 (strict) ✓.

Step 4 (difference): The external file format is a *shared artifact* — both B and C must agree on it, so changing it drags every reader of that format. With data coupling the format is internal to C and the boundary is a typed value, so the change is contained. **External (rank 4) is worse than Data (rank 1) precisely because of this counting difference.**

التبرير العربي: الـ external coupling يربط B و C بصيغة ملف مشتركة، فتغيير الصيغة يكسر B ثم A ثم Top ⇒ 4 وحدات. لو صار الربط data coupling (بارامتر/قيمة مُرجَعة ثابتة)، التغيير داخل C فقط ⇒ وحدة واحدة (أو 2 بالعدّ المتحفظ). فرق العدّ (4 مقابل 1) يشرح لماذا External أسوأ من Data على سلم Fig 7.

القسم ز — Translation into Ladder Position (رتبة الاقتران/التلاحم) + تمارين إضافية

Q7.14 (Translate a description into the correct ladder rank — combined)

> Translate each description into (a) the coupling type and its rank on Fig 7, or (b) the cohesion type and its rank on Fig 11.
>
> 1. "Module X sends module Y the value count only." — coupling?
> 2. "Module X sends module Y the entire 40-field PersonRecord though Y needs only name." — coupling?
> 3. "Module X passes isLastRecord so Y knows whether to exit its loop." — coupling?
> 4. "Module X assumes Y reads from a fixed-format `.dat` file agreed between two teams." — coupling?
> 5. "Module X changes a variable inside module Y." — coupling?
> 6. A routine whose elements are: "open log; open database; open socket" — cohesion?
> 7. A routine whose elements are: "all operations that update either the name OR the address OR the phone, chosen by a flag" — cohesion?
> 8. A routine that performs a single well-defined task. — cohesion?

Worked answer — Step 1 (map each to the ladder):

#	Description keyword	Type	Fig	rank
1	one primitive value passed	**Data** coupling	Fig 7	1 (best)
2	whole structure despite needing one field	**Stamp** coupling	Fig 7	2
3	a **flag** controlling the other's logic	**Control** coupling	Fig 7	3
4	bound to an **external file format**	**External** coupling	Fig 7	4
5	changing another module's **internals**	**Content** coupling	Fig 7	6 (worst)
6	"all at startup / same time"	**Temporal** cohesion	Fig 11	5
7	"OR … chosen by a flag"	**Logical** cohesion	Fig 11	6
8	one task, one verb	**Functional** cohesion	Fig 11	1 (best)

Step 2 (re-check positions against the verified orderings): - Coupling order = Data > Stamp > Control > External > Common > Content. #1 Data = 1 ✓; #2 Stamp = 2 ✓; #3 Control = 3 ✓; #4 External = 4 ✓; #5 Content = 6 ✓. (Common was not probed here; it sits at rank 5 — untouched.) - Cohesion order = Functional > Sequential > Communicational > Procedural > Temporal > Logical > Coincidental. #8 Functional = 1 ✓; #6 Temporal = 5 ✓; #7 Logical = 6 ✓.

Step 3 (warning on #4): Do NOT call #4 "Data coupling". The word "file" is not automatically external coupling — it becomes **External** only when the module is bound to a *format/interface imposed from outside the software* (a format two teams agreed on). Passing a file handle as a value would be Data; being tied to an externally-fixed format is External.

التبرير العربي: كل وصف يُترجم مباشرة لرتبة: تمرير قيمة مفردة = Data (1)؛ تمرير بنية كاملة = Stamp (2)؛ تمرير flag = Control (3)؛ الارتباط بصيغة ملف خارجية = External (4)؛ التعديل داخل وحدة ثانية = Content (6). تلاحم: "وقت واحد/startup" = Temporal (5)؛ "أو حسب flag" = Logical (6)؛ مهمة واحدة = Functional (1). الرتب مطابقة تمامًا لـ Fig 7 و Fig 11.

Q7.15 (Apply coupling rank to choose a refactor order — counting application)

> A design has four modules with the following problems: - **U1** is **Stamp**-coupled to its caller (passes a whole struct). - **U2** is **Content**-coupled to a neighbour (writes its internals). - **U3** is **Common**-coupled to three others (shared global). - **U4** is **Control**-coupled to its caller (passes a flag).

Task: Rank U1–U4 by refactoring urgency using the **verified** coupling order. Justify each rank with the ladder position.

Worked answer — Step 1 (place each on Fig 7):

Module	Coupling	Fig 7 rank (1=best,6=worst)
U2	Content	6
U3	Common	5
U4	Control	3
U1	Stamp	2

Step 2 (urgency = worst first): 1. **U2 (Content, rank 6)** — worst; hidden bidirectional dependency, invisible in compilers. 2. **U3 (Common, rank 5)** — shared global; change ripples to all 3 sharers. 3. **U4 (Control, rank 3)** — flag passed; couples the caller to the callee's internals logic. 4. **U1 (Stamp, rank 2)** — mildest; over-exposure but still a one-directional data pass.

Step 3 (re-check — strict monotonicity): Content(6) > Common(5) > Control(3) > Stamp(2) — urgency order U2 > U3 > U4 > U1 ✓, matching the verified order Data > Stamp > Control > External > Common > Content.

التبرير العربي: الترتيب الرقمي على Fig 7: 6 = Content (الأسوأ) ← 5 = Common ← 3 = Control ← 2 = Stamp. فالاستعجال: U2 أولًا (content)، ثم U3 (common، الـ global ينتشر لثلاث وحدات)، ثم U4 (control)، وأخيرًا U1 (stamp، الأخفّ). الترتيب مطابق حرفيًا للترتيب المُتحقق منه.

Q7.16 (Apply cohesion rank to choose which routine to split — counting)

> A "reports" module contains five routines grouped as follows: R1 = compute totals (single task); R2 & R3 = read then format (output → input chain); R4 & R5 = "all routines that either print OR export, selected by a flag". A reviewer wants to split **one** group.
>
> **Task:** Assign the cohesion rank to each group, then state which group to split first and why (using the rank numbers).

Worked answer — Step 1:

Group	Structure	Cohesion	Fig 11 rank
R1	one task	Functional	1
R2+R3	read → format chain	Sequential	2

Group	Structure	Cohesion	Fig 11 rank
R4+R5	print OR export by flag	Logical	6

Step 2 (which to split): The **R4+R5 group** (Logical, rank 6). It is the only group far from the best end; the "OR + flag" structure is exactly the Logical-cohesion marker, and it forces control coupling on the caller.

Step 3 (re-check the ranks against Fig 11): Functional = 1 ✓; Sequential = 2 ✓; Logical = 6 ✓ (order Functional > Sequential > Communicational > Procedural > Temporal > Logical > Coincidental).

Step 4 (counting consequence): Splitting R4+R5 into two functional routines (print-function, export-function) moves 2 routines from rank 6 → rank 1, and lets the caller drop its flag parameter (Control → Data coupling). Splitting R2+R3 would move 2 routines from rank 2 → rank 1 — a much smaller gain. **Urgency is proportional to the rank distance from the best level.**

التبرير العربي: R1 = Functional (1)، R2+R3 = Sequential (2)، R4+R5 = Logical (6). نُقسّم مجموعة R4+R5 أولًا لأنها الأبعد عن الأفضل ("أو" + "flag" = علامة logical cohesion)، فيرفع تلاحمها من 6 إلى 1 ويسمح للـ caller بإسقاط الـ flag (control ← data). الفرق (6←1) أكبر بكثير من (2←1)، فالمكسب أكبر.

Q7.17 (Full worked application — compute fan-in/out, classify, score, and choose a refactor)

> A structure chart: - Main calls Load, Process, Report - Load calls ReadFile, Validate - Process calls Validate, Compute, Transform, Sort, Filter, Aggregate, Rank, FormatOut, Persist (9 calls) - Report calls FormatOut - Validate is also called by Load and Process; FormatOut by Process and Report.

Task: (a) Compute every fan-out and flag breaches. (b) Compute fan-in of Validate and FormatOut. (c) Process is internally a "flag dispatcher" reading a mode parameter and is **Content**-coupled to Compute. Classify its cohesion/coupling and compute its DQS. (d) Name the two most urgent actions.

Worked answer — (a) fan-out:

Module	Calls	fan-out
Main	Load, Process, Report	3
Load	ReadFile, Validate	2
Process	Validate, Compute, Transform, Sort, Filter, Aggregate, Rank, FormatOut, Persist	**9**
Report	FormatOut	1
others	—	0

Breach: Process = 9 > 7 → **flagged ✓**. (Re-count: Validate, Compute, Transform, Sort, Filter, Aggregate, Rank, FormatOut, Persist = 9 ✓.)

(b) fan-in:

Fan-in(Validate): called by Load and Process → **2 ✓**
Fan-in(FormatOut): called by Process and Report → **2 ✓**

So Validate and FormatOut are the shared leaves — highest reuse, must have the cleanest interfaces (both are also affected by Process's breach).

(c) Process classification and DQS:

Cohesion: reads mode to choose among tasks → **Logical** (rank 6).
Coupling: Content-couples to Compute (writes its internals) — and Content (rank 6) is worse than the Control coupling the mode flag would suggest, so **Content (rank 6)** governs.
DQS = (8 − 6) + (7 − 6) = 2 + 1 = **3** (very weak; theoretical worst = 2).
Re-check: 8−6 = 2 ✓; 7−6 = 1 ✓; DQS = 3 ✓.

(d) two most urgent actions: 1. **Break the Content coupling between Process and Compute** — replace internal access with a data parameter (rank 6 → 1). Highest-severity defect. 2. **Split Process** — its 9-call fan-out (> 7, the fan-out warning) and Logical cohesion (rank 6) mean it is doing too much; decomposing it into single-purpose modules raises cohesion and removes the mode flag (Control → Data).

Step (re-check arithmetic): Process fan-out = 9 ✓; Validate fan-in = 2 ✓; FormatOut fan-in = 2 ✓; DQS = 3 ✓. All confirmed.

التبرير العربي: Process هي الوحدة المريضة: fan-out = 9 (> 7 ⇒ تحذير)، logical (rank 6) احمltel بسبب flag الـ mode، واقتران content مع Compute (rank 6) ⇒ DQS = 3 (قريب من الأسوأ). الإجراءان الأهم: (1) كسر الـ content coupling (استبداله بـ data) لأنه الأخطر؛ (2) تقسيم Process لأن fan-out يتجاوز 7 وتلاحمها logical. Validate و FormatOut عندهم fan-in = 2 فهُم وحدات إعادة استخدام تحتاج واجهات نظيفة. (قاعدة fan-out>7 موسمة كـ standard concept.)

Q7.18 (Cross-check application — put one design through the whole toolkit)

> Design **Z**: 7 modules. Module `Z3` calls `Z4, Z5, Z6, Z7, Z1, Z2, Z5` (note `Z5` appears twice in the call sheet — treat each *distinct* callee once). `Z4` passes only `Z3`'s `id` back (data). `Z3` is internally a set of three routines grouped because they all run when a batch job starts. `Z6` reads and writes a global CONFIG.

Task: (i) Compute `Z3`'s correct fan-out (watch the duplicate). (ii) Classify `Z3`'s cohesion. (iii) Classify `Z6`'s coupling. (iv) State whether the design's cost is likely left of, in, or right of the Region of Minimum Cost if the ideal is ~9–11 modules.

Worked answer — (i) fan-out of `Z3`: Raw call sheet: Z4, Z5, Z6, Z7, Z1, Z2, Z5. Distinct callees = {Z4, Z5, Z6, Z7, Z1, Z2} = **6**. (The duplicate Z5 counts once.) **Re-check:** list = Z1, Z2, Z4, Z5, Z6, Z7 → 6 distinct ✓. 6 ≤ 7 → **no breach**.

(ii) Cohesion of `Z3`: three routines grouped because they all run at batch start → **Temporal** cohesion (rank 5).

(iii) Coupling of `Z6`: shares global CONFIG with others → **Common** coupling (rank 5).

(iv) Region of Minimum Cost: the design has **7 modules** but the ideal is **9–11** → 7 < 9, so the design sits **left of the Region of Minimum Cost** (too few modules). It is under-modularised: fewer boundaries means less integration overhead but larger, harder-to-understand modules — cost is on the *left* branch of Fig 4; adding a few modules would reduce total cost.

Step (re-check): distinct fan-out = 6 ✓; Temporal rank 5 ✓; Common rank 5 ✓; 7 < 9 → left branch ✓.

التبرير العربي: الـ fan-out الصحيح = 6 لأن Z5 مكررة في كشف النداء وتُحسب مرة واحدة (لا يتجاوز 7 ⇒ لا تحذير). تلاحم Z3 = Temporal (كلها تبدأ وقت واحد، rank 5). اقتران Z6 = Common (global CONFIG، rank 5). بعدد 7 وحدات مقابل الأمثل 9–11، التصميم **يسار** منطقة الكلفة الدنيا (وحدات قليلة جدًا) فكلفته على الفرع الأيسر من منحنى Fig 4.

خريطة التغطية والتحقق (Verification Map & Coverage)

Q	نوع التطبيق	المفاتيح	المصدر
Q7.1	fan-in/out حساب	fan-out=8 breach + fan-in	s43 (Fig 18) + standard concept
Q7.2	fan-out حساب	fan-out=10 breach	standard concept
Q7.3	fan-in/out + reuse	fan-in=3 hotspot	standard concept
Q7.4	تصنيف + DQS	4 وحدات + avg 8.0	Fig 7/11 (s19, s27)
Q7.5	تصنيف + DQS	5 وحدات + avg 7.4	Fig 7/11
Q7.6	Region of Min Cost	min=9، نطاق 7–11	Fig 4 (s14)
Q7.7	Region of Min Cost	min=10، حدّة الفرعين	Fig 4 (s14)
Q7.8	one-sentence cohesion	6 دوال ← رتب	definitions (s27–30)
Q7.9	OOD 6 خطوات	قائمة مخرجات	s73–80
Q7.10	OOD 6 خطوات	مخرجات + قرارات	s73–80
Q7.11	trace/count data vs content	1 مقابل 3/6	s19–23
Q7.12	trace/count common vs data	5 مقابل 1	s21
Q7.13	trace/count external vs data	4 مقابل 1	s23
Q7.14	رتبة ladder	8 وصف ← رتب	Fig 7/11
Q7.15	ترتيب إصلاح بالرتب	U2>U3>U4>U1	Fig 7
Q7.16	ترتيب إصلاح بالرتب	R4+R5 أولًا	Fig 11
Q7.17	شامل (DQS+class+fan)	Process DQS=3	Fig 7/11 + standard
Q7.18	شامل + duplicate trap	fan-out=6، يسار المنطقة	Fig 4/7/11

فهرس الاقتران المُستعمل (تحقق): (1)Data < (2)Stamp < (3)Control < (4)External < (5)Common < (6)Content — مطابق Fig 7 (slide 19). **فهرس التلاحم المُستعمل (تحقق):** (1)Functional < (2)Sequential < (3)Communicational < (4)Procedural < (5)Temporal < (6)Logical < (7)Coincidental — مطابق Fig 11 (slide 27).

ملاحظات عدم يقين (declared — Uncertainties)

1. **fan-out > 7 و reuse = high fan-in:** سلايدات/كتاب الفصل 5 **لا ينصّان رقميًا** على حدّ fan-out ولا على قاعدة الـ fan-in؛ هي قواعد إرشادية متعارف عليها في هندسة البرمجيات. سلايدات الـ structure chart (Fig 17–18، سلايدات 42–43) تعرض الرموز فقط. **وُسمت في كل موضع (Q7.1, Q7.2, Q7.3, Q7.17).**

2. **الترتيب العددي (1 ranks 6–1 و7–7) و DQS:** الترتيب **مطابق** للفصل (Fig 7, Fig 11)، لكن تحويله إلى أرقام وحساب DQS = (8−r)+(7−r) هو **اصطلاح مخترع لهذا الملف**، مُعلّن في "قاعدة العمل" وفي كل تمرين.
3. **أرقام منحنى الكلفة (Q7.6, Q7.7):** slide 14 هو **شكل فقط** ("Fig. 4: Modularity and software cost") بلا أرقام؛ الجداول الرقمية **توضيحية من صناعة هذا التمرين** لتمثيل شكل U والـ Region of Minimum Cost، **مو من الفصل**. المنطقة تُقاس بنسب شكل المنحنى لا بأرقام الكتاب.
4. **"Design Quality Score" و"Average DQS":** مفهوم تقييمي مخترع للتمرين؛ الفصل يعطي معيار max cohesion/min coupling نصًّا ولا يعطي معادلة نقاط.
5. **عدد مستويات التلاحم:** slide 26 يسرد **6** أنواع (بدون Communicational)، بينما slide 27 (Fig 11) والتعريفات تعرض **7** مع Communicational. الملف يعتمد الـ **7** (متوافق مع VERIFIED FACTS و Fig 11).
6. **Any "if required" تلاحم/خطوة:** خطوتا OOD (activity، state chart) موسومتان "If required" في slide 73 و 79 — عُوملتا كاختياريتين في Q7.9/Q7.10.
7. **كلفة left/right branch on the (Q7.18):** التصنيف "left of the region" مبني على مقارنة عدد الوحدات (7) بالمدى الأمثل المعلّن (~9–11) — **استنتاج من شكل Fig 4** لا نصّ حرفي.

المصدر: Singh & Aggarwal, Software Engineering 3rd ed., Ch.5 — سلايدات 1–99 (Fig 4 ص14، Fig 7 ص19، Fig 11 ص27، خطوات OOD ص73–80) + book p.202–251. كل رتبة قوبلت مع Fig 7/Fig 11، وكل حساب fan-in/out وDQS أُعيد حسابه مرتين قبل التثبيت.

Retrieval set (active recall)

السؤال	الجواب
ترتيب الاقتران (أفضل ← أسوأ)؟	Data · Stamp · Control · External · Common · Content
ترتيب التلاحم (أفضل ← أسوأ)؟	Functional · Sequential · Communicational · Procedural · Temporal · Logical · Coincidental
شنو DQS؟	درجة جودة التصميم (Design Quality Score)
شنو Region of Minimum Cost؟	المنطقة المثلى على منحنى المعيارية–الكلفة